

University of Gujrat
CS-464 Human Computer Interaction
Department of Computer Science
Faculty of Computing & Information Technology

Title	Human Computer Interaction
Code	CS-464
Credit Hours	3 Theory/week: Weight 3 Cr. Hrs. Lectures: 2 Duration 1.5 Hrs. /Lecture
Prerequisite	None
Follow Up	Advanced UI/UX Design, Software Engineering
Category	Domain Elective
Aims and Objectives	This course is designed to provide students with a comprehensive introduction to the field of Human Computer Interaction (HCI). The course aims to impart foundational knowledge required to design useful and usable interactive systems based on users' needs and contextual requirements. It focuses on understanding human capabilities, interaction models, design principles, usability evaluation techniques, and the development of prototypes to enhance user experience.
Learning Outcomes	At the end of the course the students will be able to: CLO 1: Understand humans, the computer, and their interaction mechanisms. (C2 – Understand) CLO 2: Apply interaction design principles, design rules, and cognitive models in system development. (C3 – Apply) CLO 3: Analyze socio-organizational issues, stakeholder requirements, and collaboration models in system design. (C4 – Analyze) CLO 4: Evaluate interactive systems using usability and evaluation techniques. (C3 – Apply)
Course	The course will include theoretical concepts of Human Computer Interaction and practical sessions involving: • Wireframing • Prototyping • Usability Testing • User Evaluation Techniques

Reference Material	A. Dix, J. Finlay, G. Abowd and R. Beale. Human Computer Interaction, Prentice Hall (Latest Edition) J. Preece, Y. Rogers, H. Sharp. Interaction Design: Beyond Human-Computer Interaction, 6th Edition, Wiley 2023								
Assessment Criteria		Sessional	25%	Mid Term	25%	Final Term	50%	Total	100%
		Quizzes	10						
		Assignments	10	25		50			
		Presentations	5						

Framework				
Week	Lecture	Topic	Source (Book-Chapter No. Section No.)	Recommendations for Learning Activities (Mention Assignments, Test, Quizzes, Practical, Case Study, Projects, Lab Work or Reading Assignments)
1	1	Introduction to the fields of HCI, The Human Factor		Distribution of course outline
	2			
2	3	The Computer Factor		
	4	The Interaction,		
3	5	Models of Interaction,		
	6	Interaction Paradigms		
4	7	Human psychology and ergonomics,		
	8	HCI in the Software Process		
5	9	Interaction Design basics and the design process		
	10			Quiz 1
6	11	Design Principles, Principles to support usability		
	12	Flexibility and Robustness		Assignment 1
7	13	Principles of learnability		
	14	Flexibility and Robustness		
8	15	Shneider man's Eight golden Rules		
	16	Nielsen's heuristics		

Mid-Term Exam					
9	17	Introduction to Evaluation, Evaluation Techniques			
	18				
10	19				
	20	Experts Evaluations,			
11	21	Cognitive Walkthrough,			
	22	Heuristic Evaluation,			
12	23	Evaluating through user participation		Quiz 2	
	24	Laboratory Evaluation			
13	25	Field Evaluation,			
	26	Observational Techniques			
14	27	Query techniques and Think-Aloud Techniques,			
	28	Usability Testing,		Assignment 2	
15	29	Contemporary topics and issues			
	30				
16	31	Conducting Experimental,			
	32	Understanding the process of A/B testing			
* Some Advanced Topics may be Discussed or Given as an Assignments					
Final-Term Exam					